

CYCLOSILICATES

BENITOITE

Hexagonal

H 6–6½ • SG 3.64–3.65

This usually blue barium titanium silicate occurs in serpentinite and veins in schist. Gem-quality crystals come from California, USA.



— six-sided crystal

TOURMALINE

Trigonal

H 7 • SG 2.9–3.1

Tourmaline is the name for a group of 11 hydrous boron silicate minerals with the same crystal structure but varying chemistry.



— prismatic crystal

AQUAMARINE

EMERALD

BERYL

Hexagonal

H 7½–8 • SG 2.63–2.92

This beryllium aluminium silicate is both a source of beryllium and a gemstone. Gem varieties include emerald (green) and aquamarine (greenish blue).

**SUGILITE**

Hexagonal

H 6–6½ • SG 2.74–2.79

This rare silicate of potassium, sodium, iron, and lithium occurs in metamorphic rock.

**MORGANITE**

Hexagonal

H 7½–8 • SG 2.63–2.92

Morganite is a pink variety of beryl, coloured by additional cesium or manganese. It forms tabular crystals in pegmatites.

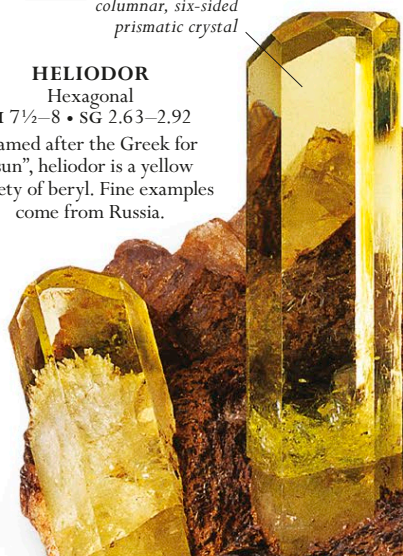
— columnar, six-sided prismatic crystal

HELIODOR

Hexagonal

H 7½–8 • SG 2.63–2.92

Named after the Greek for “sun”, heliodor is a yellow variety of beryl. Fine examples come from Russia.



— rock groundmass

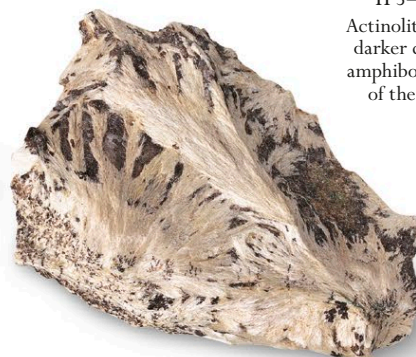
INOSILICATES

ACTINOLITE

Monoclinic

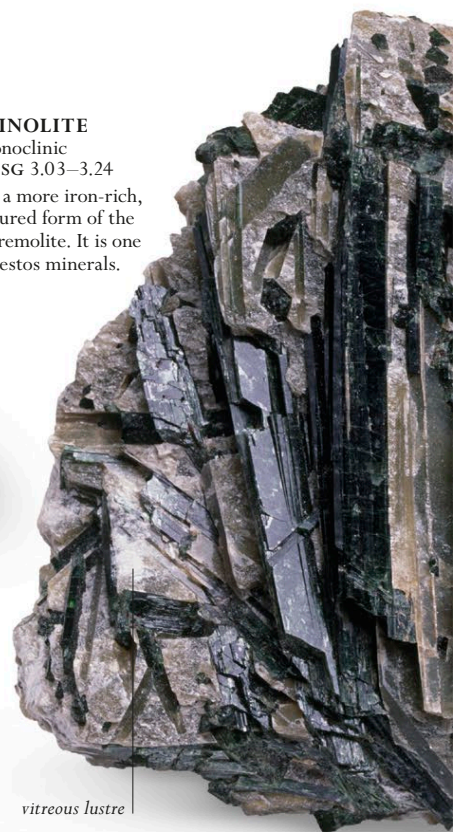
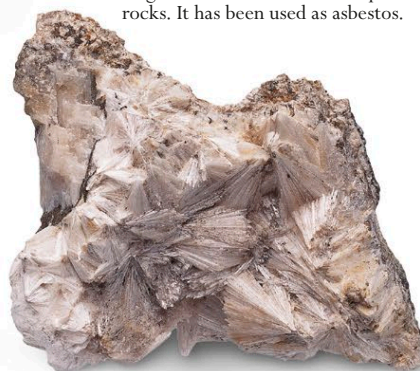
H 5–6 • SG 3.03–3.24

Actinolite is a more iron-rich, darker coloured form of the amphibole tremolite. It is one of the asbestos minerals.

**TREMOLITE**

Monoclinic • H 5–6 • SG 2.99–3.03

A widespread amphibole, this hydrous silicate of calcium and magnesium forms in metamorphic rocks. It has been used as asbestos.



— vitreous lustre

**NEPHRITE**

Monoclinic • H 6½ • SG 2.99–3.24

This very tough, cream to dark green form of the amphiboles tremolite and actinolite is commonly known as jade.

PECTOLITE

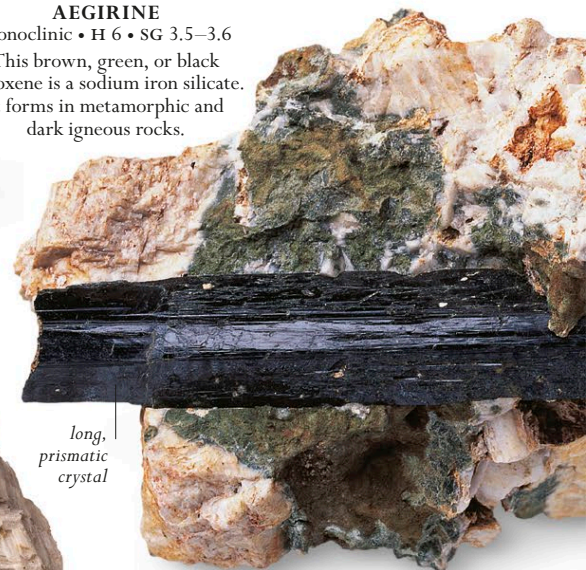
Triclinic • H 4½–5 • SG 2.48–2.9

This sodium calcium silicate hydroxide forms in cavities within basalt. It is common in Canada, the USA, and England.

**AEGRINE**

Monoclinic • H 6 • SG 3.5–3.6

This brown, green, or black pyroxene is a sodium iron silicate. It forms in metamorphic and dark igneous rocks.



— long, prismatic crystal

HORNBLende

Monoclinic • H 5–6 • SG 3–3.4

Common in igneous and metamorphic rocks, this amphibole mineral is a dark, hydrous silicate of calcium, magnesium, iron, and aluminium.



— fibrous mass

WOLLASTONITE

Triclinic • H 4½–5 • SG 2.86–3.09

This calcium silicate, found in marble and other metamorphic rocks, is used in ceramics, paints, and as a form of asbestos.

RHODONITE

Triclinic

H 5½–6½ • SG 3.57–3.76

Rose-red or pink rhodonite is a manganese calcium silicate, occurring as crystals, masses, and grains. It is used in jewellery making.

